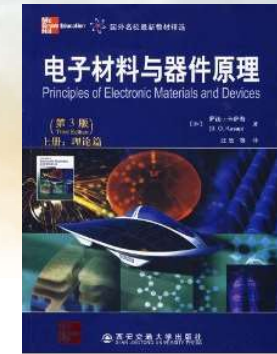
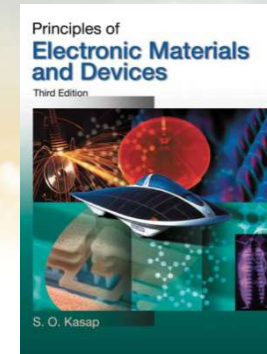




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch2-4 Electrical & Thermal Conduction in Solids

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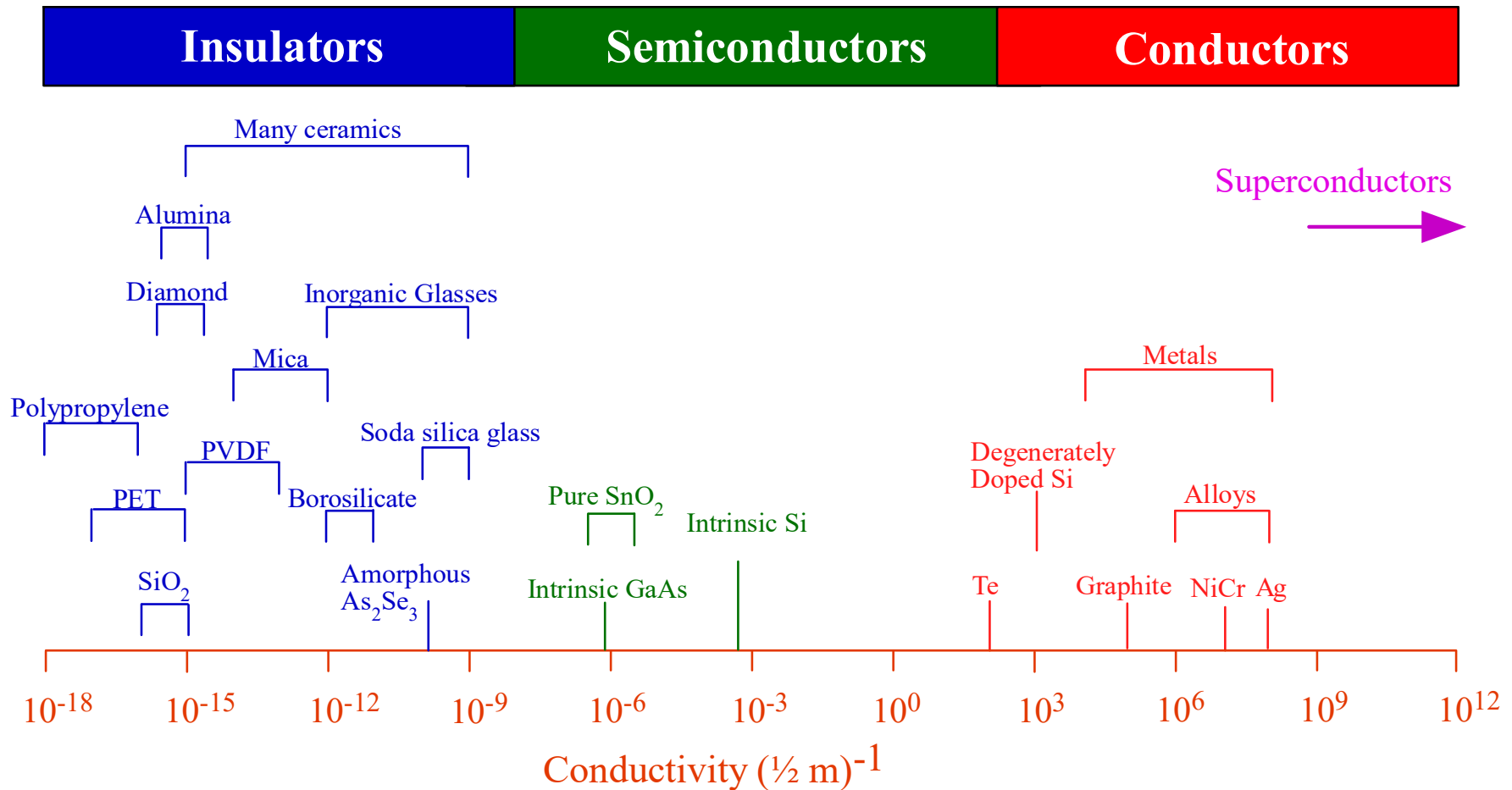
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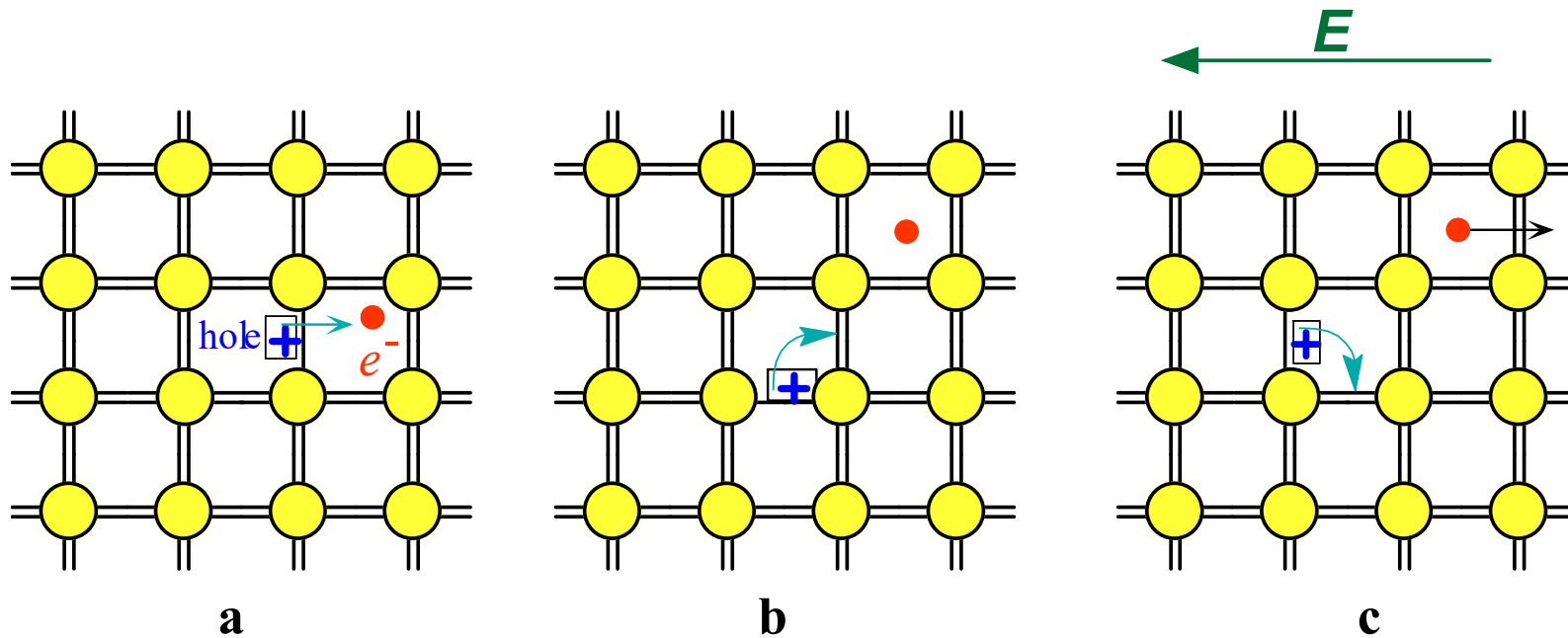


Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

2.7 Electrical Conductivity of Nonmetals (非金属的电导率)



2.7.1 Semiconductor (半导体)

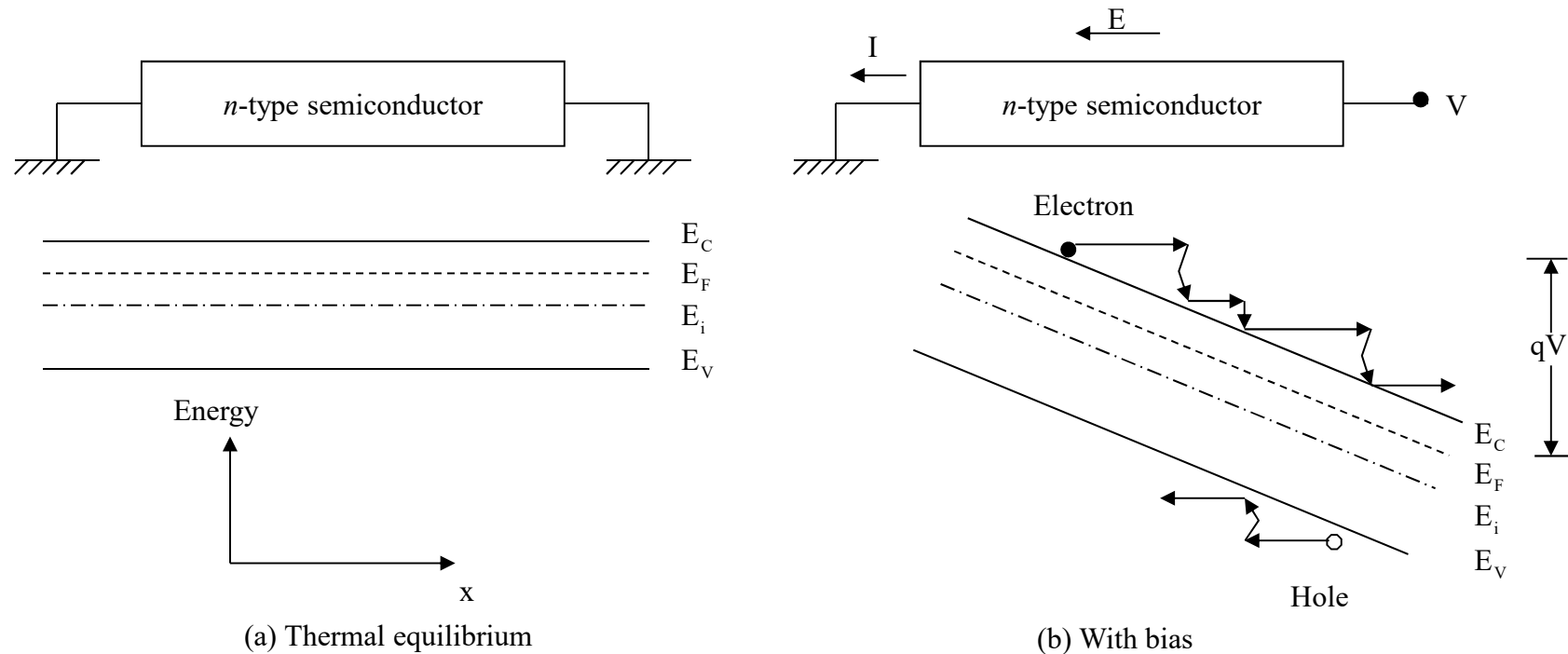


- (a) Thermal vibrations of the atoms rupture a bond and release a free electron into the crystal. A hole is left in the broken bond which has an effective positive charge.
- (b) An electron in a neighboring bond can jump and repair this bond and thereby create a hole in its original site; the hole has been displaced.
- (c) When a field is applied both holes and electrons contribute to electrical conduction.



Energy band under e -field: drift of carriers

Consider electrical conduction in a uniform semiconductor. Figure (a) shows an n -type semiconductor and its energy band diagram in thermal equilibrium. (b) A band diagram corresponding to the material with bias. Assume that the contact for both cases are Ohmic contacts.

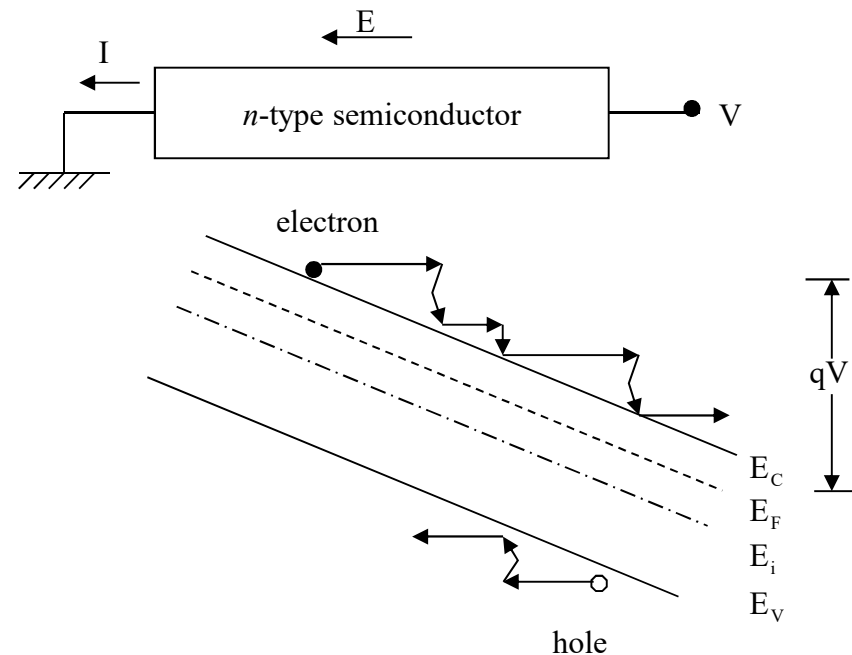


The Drift of Carriers (载流子漂移)

Electrons in the conduction band move to the right under the bias. The kinetic energy is equivalent to the distance at the edge of the band (E_C for electrons). When an electron experiences a collision, it will lose some or even all of the kinetic energy. It will dissipate into the crystal lattice and returned to the position at thermal equilibrium. Same process will be repeated many times. The conduction of holes can also be imagined in a similar manner, but in opposite directions.

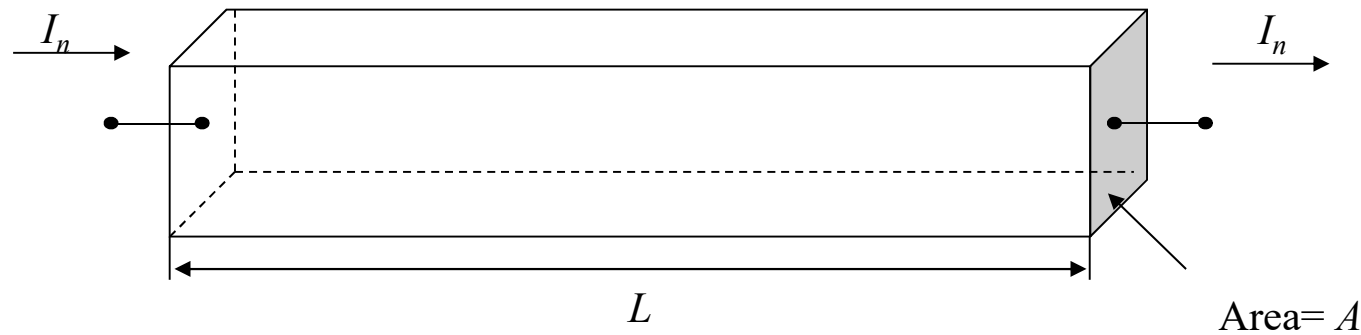
With the applied electric field, the transport of carriers generates a current called a **drift current**.

(漂移电流)



The Drift of Carriers (载流子漂移)

Consider a semiconductor with a cross-section of A , length of L , and a carrier concentration of n electrons per cubic centimeter, as shown in the figure



Assuming that an electric field E is applied to the sample, the electron current density J_n flowing through the sample is equal to the sum of the electron charge density ($-e$) multiplies the electron velocity, ie

$$J_n = \frac{I_n}{A} = \sum_{i=1}^n (-ev_i) = -env_n = en\mu_n E$$

where I_n is the current. The above formula utilize $v_n = -\mu_n E$



The Drift of Carriers (载流子漂移)

Similar results are obtained for holes, but the charge carried by the holes is positive.

$$J_p = epv_p = ep\mu_p E$$

Therefore, the total current flowing through the semiconductor due to the applied electric field is the sum of the electron and hole currents, ie

$$J = J_n + J_p = (en\mu_n + ep\mu_p)E = \sigma E$$

Therefore, the resistivity is

$$\rho \equiv \frac{1}{\sigma} = \frac{1}{e(n\mu_n + p\mu_p)} \quad \left\{ \begin{array}{ll} \rho = \frac{1}{en\mu_n} & \text{for } n\text{-type semiconductors,} \\ & (n \gg p) \\ \rho = \frac{1}{ep\mu_p} & \text{for } p\text{-type semiconductors,} \\ & (p \gg n) \end{array} \right.$$

The overall conductivity : $\sigma = ep\mu_h + en\mu_e$

hole (h^+): 空穴

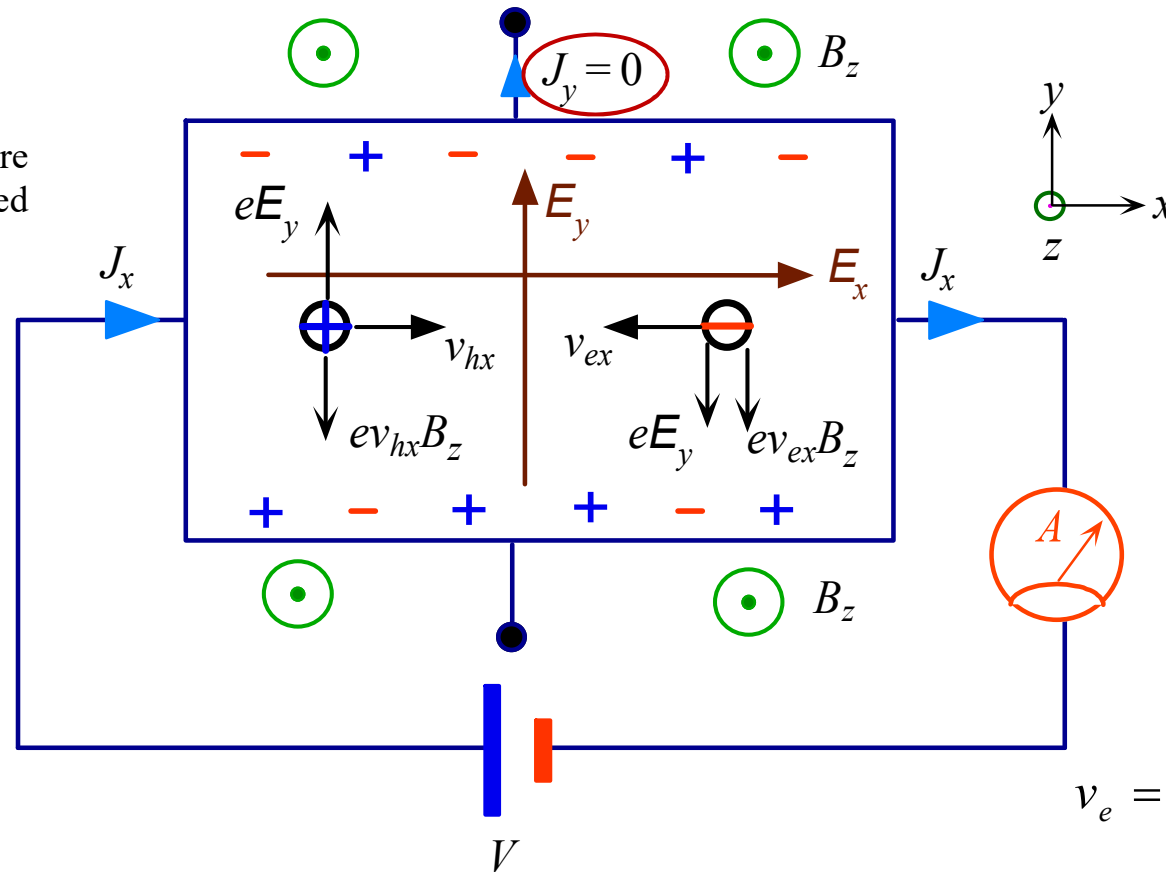
conduction electron (e^-): 导带电子



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Hall Effect in Semiconductors 半导体中的霍尔效应

Assume that more **holes** are accumulated on the bottom surface



$$a = \frac{eE}{m}$$

$$F_{\text{net}} = eE$$

$$v_e = \mu_e E = \frac{\mu_e}{e} F_{\text{net}}$$

Hall effect for ambipolar conduction as in a semiconductor where there are both electrons and holes. The magnetic field B_z is out from the plane of the paper. Both electrons and holes are deflected toward the bottom surface of the conductor and consequently the Hall voltage depends on the relative mobilities and concentrations of electrons and holes.

$$J_y = J_h + J_e = epv_{hy} + env_{ey} = 0$$



Hole

Electron

$$F_{hy} = eE_y - ev_{hx}B_z$$

$$-F_{ey} = eE_y + ev_{ex}B_z$$

$$v_e = \frac{\mu_e}{e} F_{\text{net}}$$

$$F_{hy} = \frac{ev_{hy}}{\mu_h}$$

$$-F_{ey} = \frac{ev_{ey}}{\mu_e}$$

$$\frac{ev_{hy}}{\mu_h} = eE_y - ev_{hx}B_z$$

$$\frac{ev_{ey}}{\mu_e} = eE_y + ev_{ex}B_z$$

$$\frac{v_{hy}}{\mu_h} = E_y - v_{hx}B_z$$

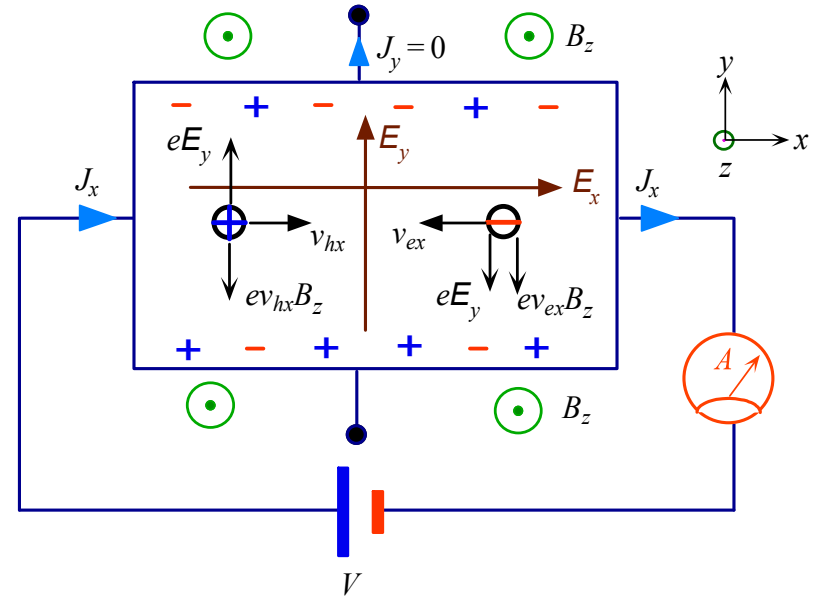
$$\frac{v_{ey}}{\mu_e} = E_y + v_{ex}B_z$$

$$v_{hx} = \mu_h E_x \quad v_{ex} = \mu_e E_x$$

$$epv_{hy} + env_{ey} = 0$$

$$p\mu_h E_y - p\mu_h^2 E_x B_z = -n\mu_e E_y - n\mu_e^2 E_x B_z$$

$$E_y (p\mu_h + n\mu_e) = B_z E_x (p\mu_h^2 - n\mu_e^2)$$



$$J_x = epv_{hx} + env_{ex} = (p\mu_h + n\mu_e)eE_x$$

$$eE_y (p\mu_h + n\mu_e)^2 = B_z J_x (p\mu_h^2 - n\mu_e^2)$$

$$R_H = \frac{E_y}{J_x B_z} \quad b = \frac{\mu_e}{\mu_h}$$

$$R_H = \frac{(p\mu_h^2 - n\mu_e^2)}{e(p\mu_h + n\mu_e)^2} = \frac{p - nb^2}{e(p + nb)^2}$$



EXAMPLE 2.22

HALL COEFFICIENT OF INTRINSIC SILICON At room temperature, a pure silicon crystal (called **intrinsic silicon**) has electron and hole concentrations $n = p = n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, and electron and hole drift mobilities $\mu_e = 1350 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_h = 450 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. Calculate the Hall coefficient and compare it with a typical metal.

SOLUTION

Given $n = p = n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, $\mu_e = 1350 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, and $\mu_h = 450 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, we have

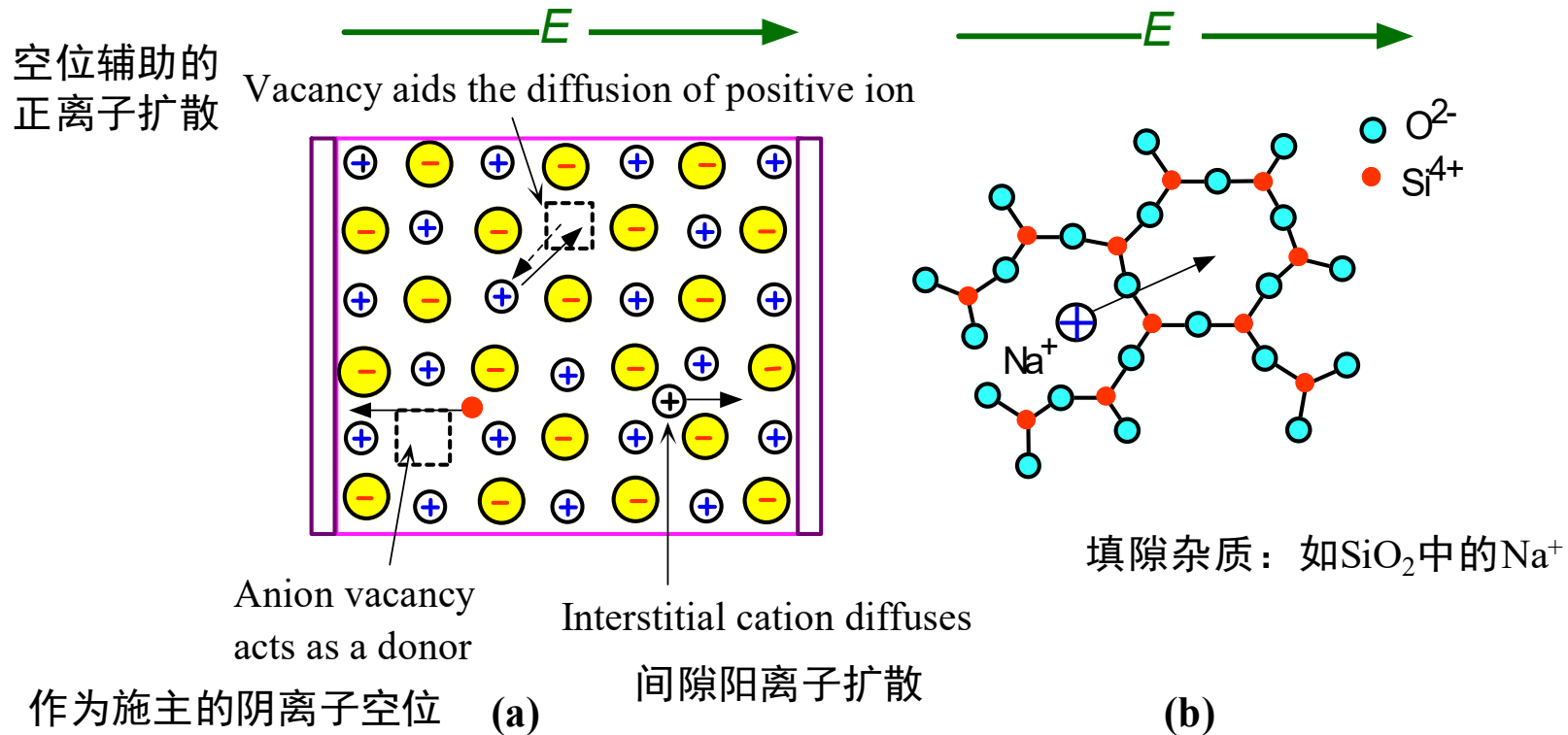
$$b = \frac{\mu_e}{\mu_h} = \frac{1350}{450} = 3$$

Then from Equation 2.50,

$$\begin{aligned} R_H &= \frac{(1.5 \times 10^{16} \text{ m}^{-3}) - (1.5 \times 10^{16} \text{ m}^{-3})(3)^2}{(1.6 \times 10^{-19} \text{ C})[(1.5 \times 10^{16} \text{ m}^{-3}) + (1.5 \times 10^{16} \text{ m}^{-3})(3)]^2} \\ &= -208 \text{ m}^3 \text{ A}^{-1} \text{ s}^{-1} \end{aligned}$$

which is orders of magnitude larger than that for a typical metal. All Hall-effect devices use a semiconductor rather than a metal sample.

2.7.2 Ionic Crystals & Glasses 离子晶体与玻璃



Possible contributions to the conductivity of ceramic and glass insulators (a) Possible mobile charges in a ceramic (b) A Na⁺ ion in the glass structure diffuses and therefore drifts in the direction of the field. (E is the electric field.)



Conductivity in Ionic Crystal

The carriers participating in the electrical conduction are ions, which are ions and vacancies. They can be classified into two categories.

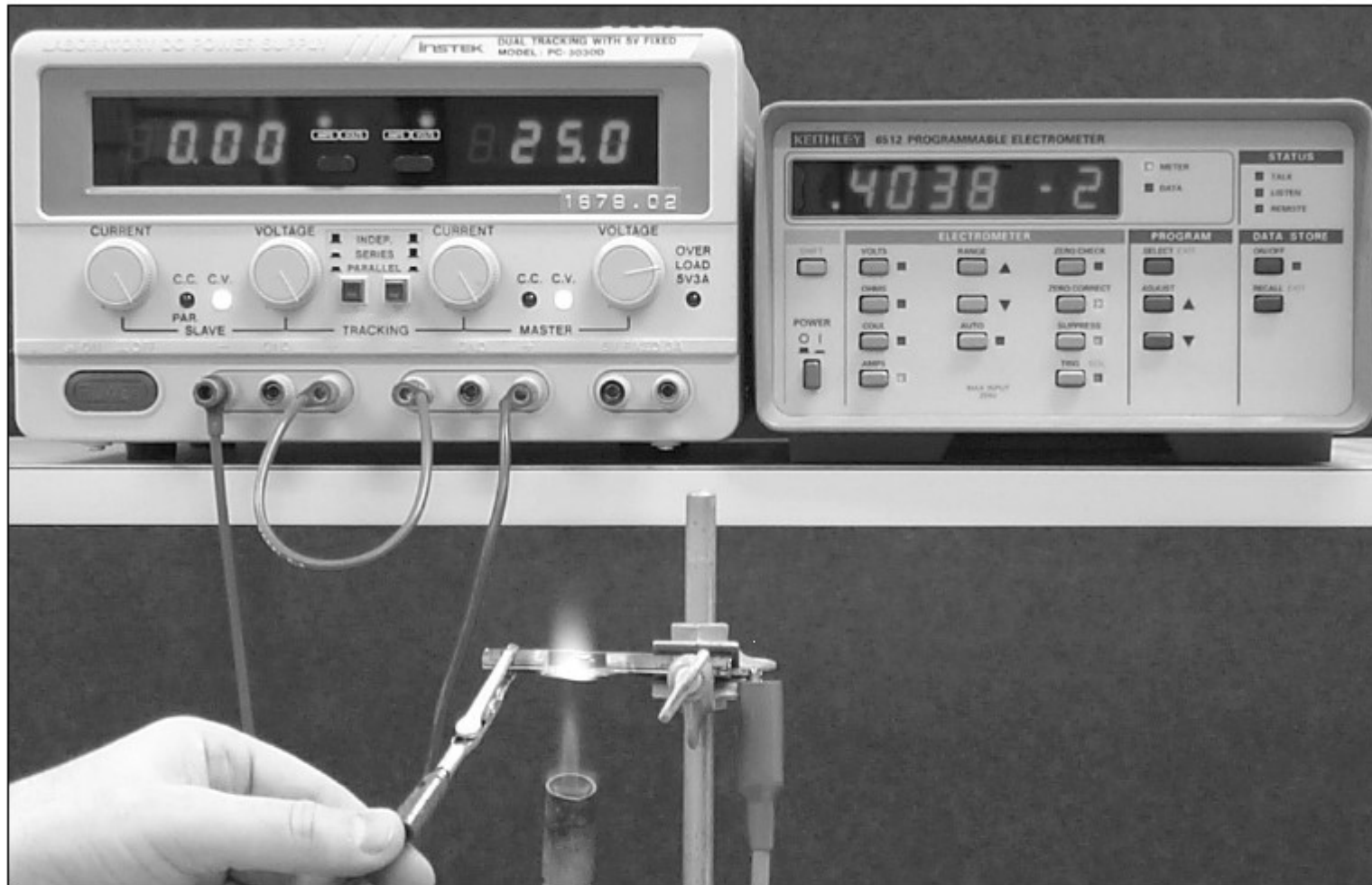
1) Intrinsic Conductivity: Movement of basic ions originating from the crystal lattice. The ions themselves leave the lattice due to thermal vibrations and form defects. This results in the generation of carriers, which are particularly pronounced at high temperatures.

- The Frenkel defect is an interstitial ion-vacancy
- Schottky defects are cationic vacancies: anion vacancy pairs

2) Impurity Conductivity: caused by the movement of weakly fixed ions (impurities). The impurity ions are weakly associated ions, their conductance shows significant at lower temperatures.

- Impurity conduction: directional movement of impurity ions
- Interstitial impurities or replacement impurities (solute)





This soda glass rod when heated under a torch becomes electrically conducting. It passes 4 mA when the voltage is 50 V (2×25 V); a resistance of 12.5 k Ω ! Ordinary soda glass at room temperature is an insulator but can be quite conducting at sufficiently high temperatures.

From *Principles of Electronic Materials and Devices, Third Edition*, S.O. Kasap (© McGraw-Hill, 2005)



Factors Influent the Conductivity

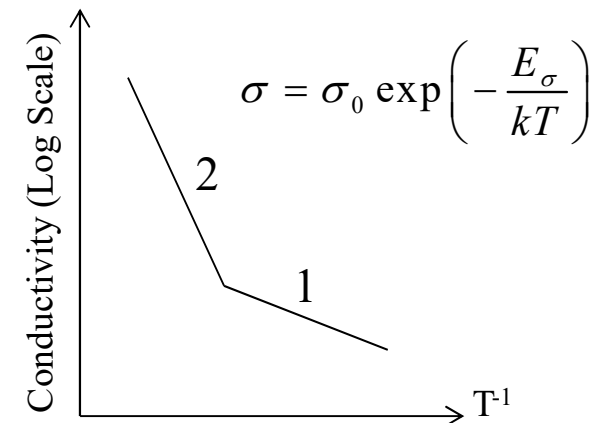
1) Temperature: Exponential relationship, and conductivity increases rapidly with temperature. Note: at low temperatures, impurity conductivity dominates (1), and at high temperatures, intrinsic conductivity plays a major role (2).

2) Crystal Structure: Activation energy is determined by the binding force of the particles between the crystals.

- a) Generally, the ionic radius is small, the binding force is large, and the activation energy is also large
- b) Ionic charge, high electricity price, large binding force, and thus high activation energy;
- c) The higher degree of accumulation, the tighter the combination, the smaller the number of ions available for movement, and the more difficult it is to move, resulting in lower conductivity, i.e. greater activation energy.

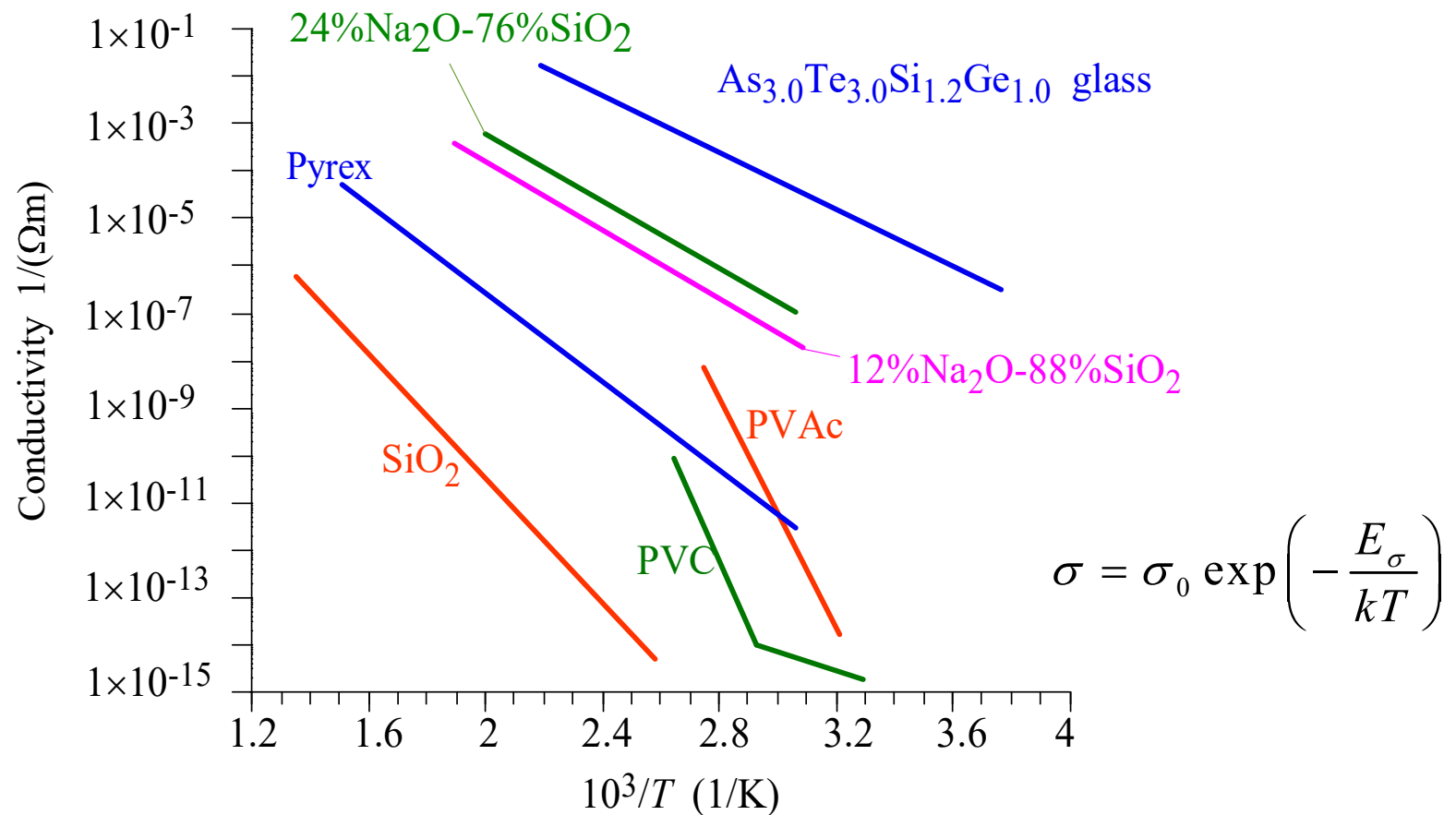
3) Crystal Defects

- a) The concentration of electron carriers is small.
- b) The ionic lattice defect concentration is large and participates in conductance. Therefore, the formation of ionic lattice defects and their concentration are the key to determine ion conductance. The main reasons for the formation and concentration of lattice defects are: thermal excitation to generate lattice defects (Schottky defects and Frenkel defects).



Conductivity v.s. reciprocal T





Conductivity vs reciprocal temperature for various low conductivity solids. (PVC = Polyvinyl chloride; PVAc = Polyvinyl acetate.) Data selectively combined from numerous sources.

The negative slope: thermal activation energy (热激活能)



EXAMPLE 2.23

CONDUCTIVITY OF A SODA-SILICATE GLASS Figure 2.29 shows the temperature dependence of 12% Na₂O–88% SiO₂, soda-silicate glass which has 12 mol% Na₂O and 88 mol% SiO₂. Calculate the activation energy of conductivity and compare this with the activation energy for the diffusion of Na⁺ ions in the soda-silicate glass structure which is in the range 0.65–0.75 eV.

SOLUTION

According to Equation 2.52 when $\ln(\sigma)$ is plotted against $1/T$, the slope should be $-E_\sigma/k$. If the conductivity at temperatures T_1 and T_2 are σ_1 and σ_2 , respectively, then the slope of the straight line for 12% Na₂O–88% SiO₂ in Figure 2.29 is

$$\text{Slope} = \frac{\ln(\sigma_2/\sigma_1)}{(1/T_2 - 1/T_1)} = -\frac{E_\sigma}{k}$$

Taking $\sigma_1 = 10^{-4} \Omega^{-1} \text{ m}^{-1}$ and $\sigma_2 = 10^{-6} \Omega^{-1} \text{ m}^{-1}$ in Figure 2.29, we find $1/T_1 = 0.00205$ and $1/T_2 = 0.00261$. Then, E_σ as eV is

$$E_\sigma = -\frac{\ln(\sigma_2/\sigma_1)}{(1/T_2 - 1/T_1)} \frac{k}{e} = -\frac{\ln(10^{-6}/10^{-4})}{(0.00261 - 0.00205)} \frac{1.38 \times 10^{-23}}{1.602 \times 10^{-19}} = 0.71 \text{ eV}$$

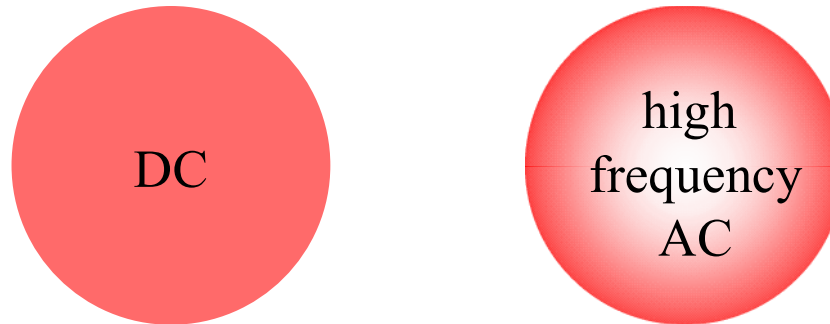
A similar calculation for the 24% Na₂O–76% SiO₂ gives an activation energy of 0.69 eV.

Both of these activation energies are comparable with the activation energy for the diffusion of Na⁺ ions in the structure. Thus, Na⁺ diffusion is responsible for the conductivity.



Additional 2.8 Skin Effect (趋肤效应)

Distribution of currents in conductor



- Skin effect is the tendency of an **alternating electric current (AC)** to become distributed within a conductor such that the current density is largest near the surface of the conductor, and decreases with greater depths in the conductor.
- The electric current flows mainly at the "skin" of the conductor, between the outer surface and a level called the skin depth.
- The skin effect causes the effective resistance of the conductor to increase at higher frequencies where the skin depth is smaller, thus reducing the effective cross-section of the conductor.
- The skin effect is due to opposing eddy currents induced by the changing magnetic field resulting from the alternating current.

